

Advanced AFM Applications Training



PeakForce-QNM I: Force Curve & PF-QNM

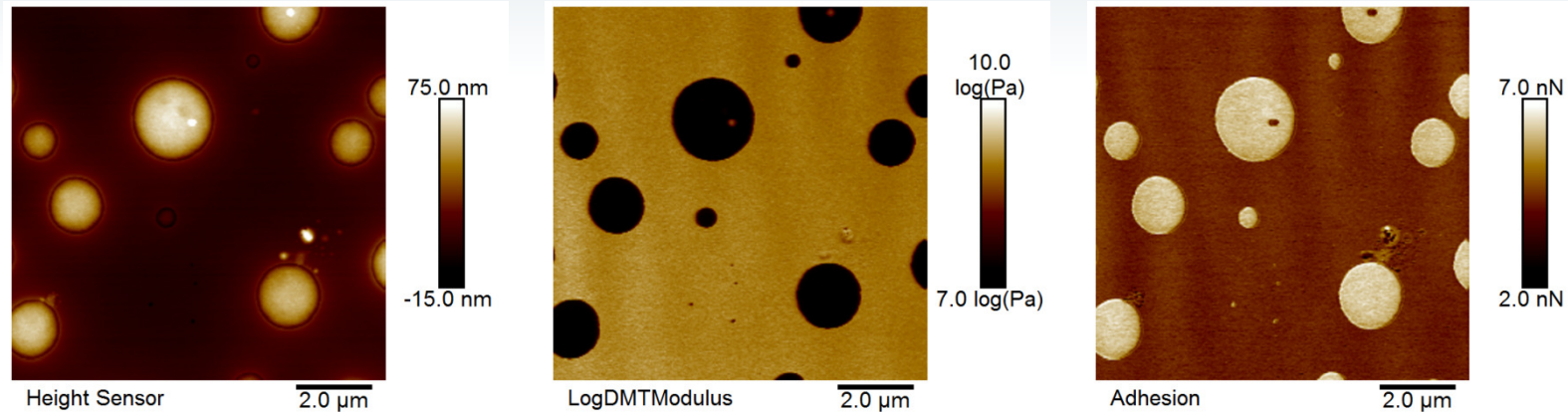
Yueming Hua, Ph.D.



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What is PeakForce QNM?

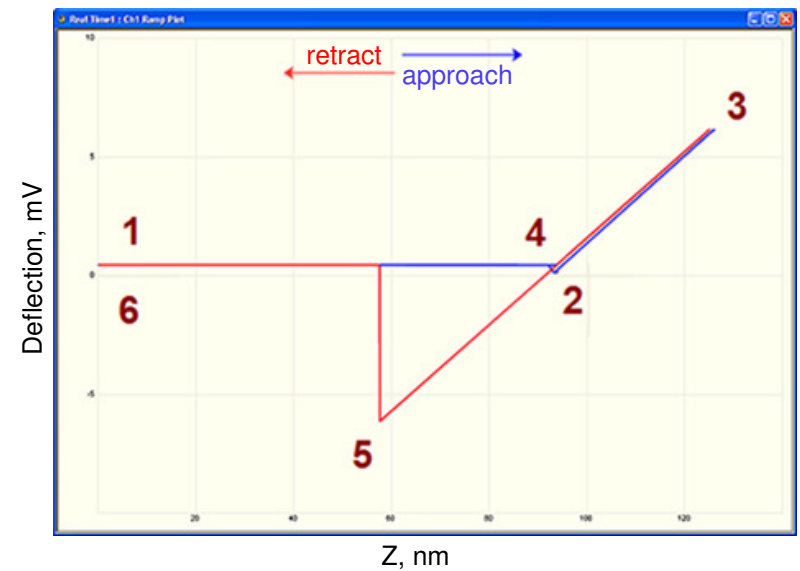
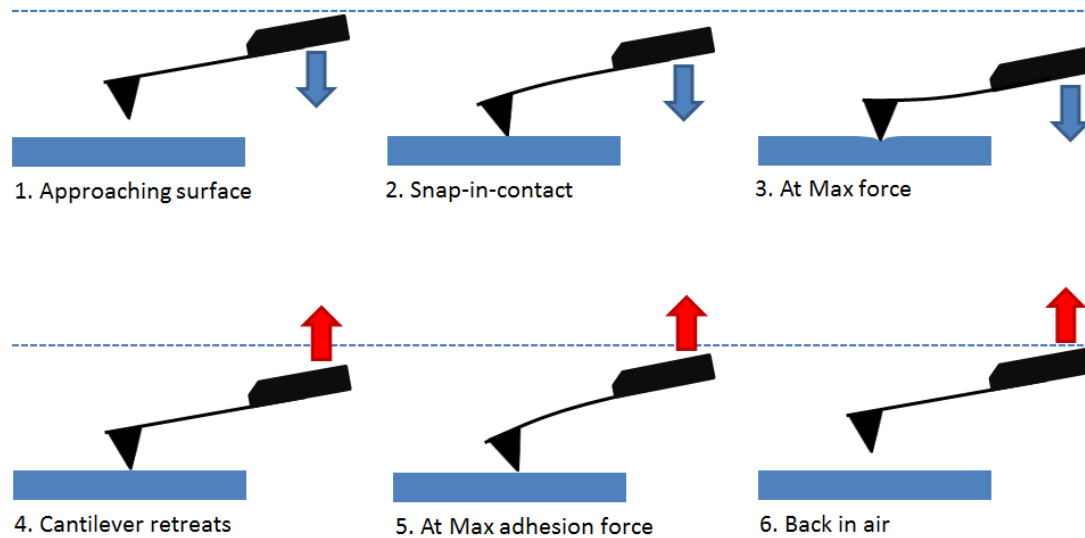


Polystyrene and Low Density Polyethylene Blend (PS-LDPE)

- PeakForce QNM is an imaging mode that produces height images and **Q**uantitative **N**ano-**M**echanical property images at the same time.
- Two parts to PeakForce QNM
 - The PeakForce Tapping part (ScanAsyst)
 - PeakForce Tapping is the feedback mode used to image the sample surface.
 - The QNM part
 - PeakForce Tapping mode produces force curves
 - The force curves are used to extract quantitative material properties data.

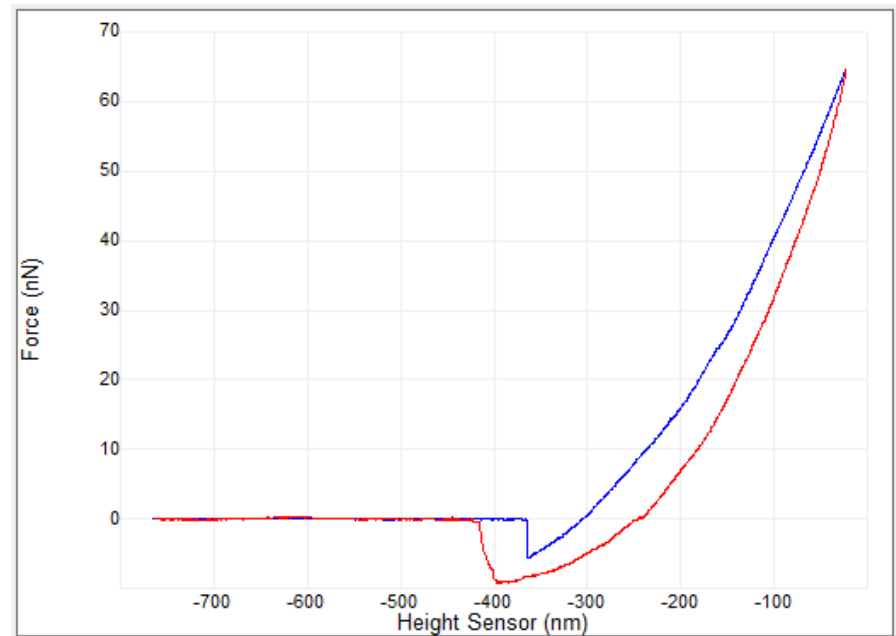
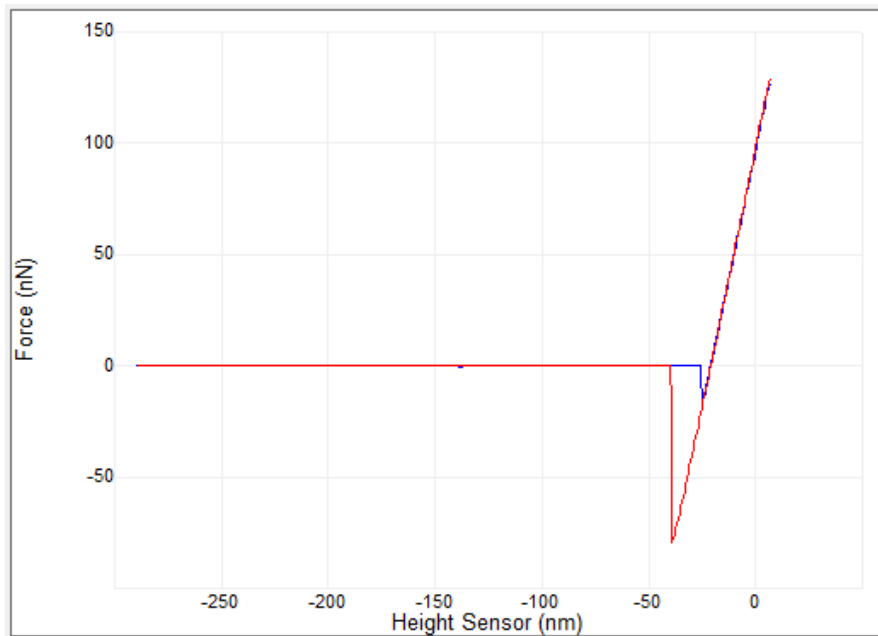
Force Curve & Contact Mechanics

Force Distance Curve



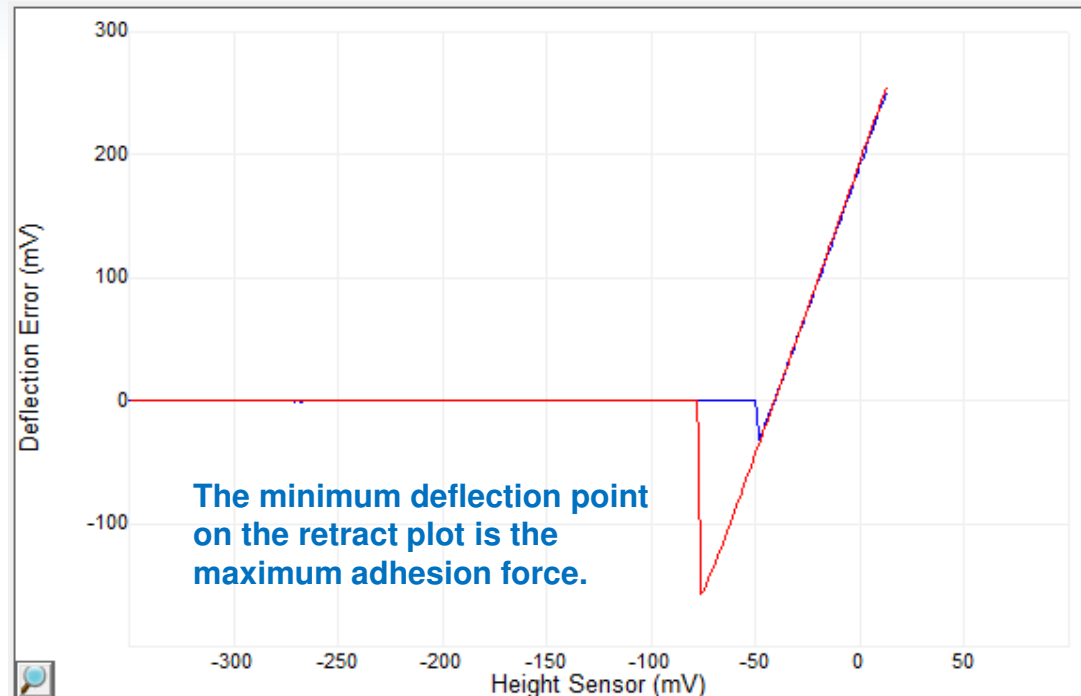
Typical steps in a standard Force-Distance curve

Force Curve Interpretation



Question: What is the difference between above two force curves?

Sample Adhesion Force



- Without calibration, ramp curve give Deflection vs. Z
- Deflection raw unit is in **V**
- How to convert Deflection from **V** to **nm**?
- How to convert Deflection in **nm** to Force in **nN**?

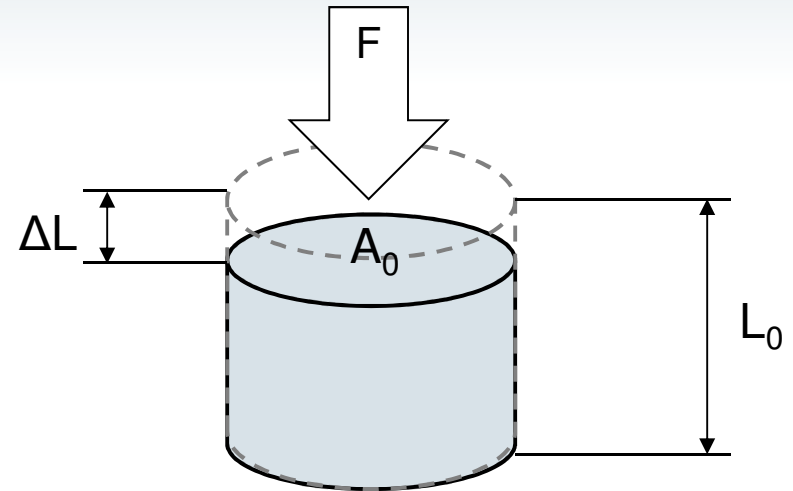
Young's Modulus



$$E = \frac{\text{tensile stress}}{\text{extensional strain}} = \frac{\sigma}{\varepsilon} = \frac{F/A_0}{\Delta L/L_0} = \frac{FL_0}{A_0\Delta L}$$

Where

E is the Young's modulus (Modulus of elasticity)
 F is the force exerted on an object under tension
 A₀ is the area where force is applied
 ΔL is the length change of the object
 L₀ is the original length of the object



Young's Modulus of Common Materials

Material	GPa	Material	GPa
Rubber	0.01–0.1	HOPG (C)	18
PTFE (Teflon)	0.5	Aluminum (Al)	69
Low density polyethylene	0.11–0.45	Titanium (Ti)	110
High density polyethylene	0.8	Copper (Cu)	117
Polystyrene	2.5–3.5	Silicon (Si)	130–185
Nylon	2–4	Tungsten (W)	400–410
Oak wood (along grain)	11	Single-walled carbon nanotube (C)	1,000
Fused silica	72	Graphene (C)	1,050
Sapphire	460	Diamond (C)	1,220

Sample Modulus: Hertzian Model



$$F = \frac{4}{3} \frac{E}{(1 - \nu^2)} \sqrt{R} \delta^{3/2}$$

or

$$(F)^{2/3} = \left(\frac{4}{3} \frac{E}{(1 - \nu^2)} \sqrt{R} \right)^{2/3} \delta$$

Linearized Equation

Where

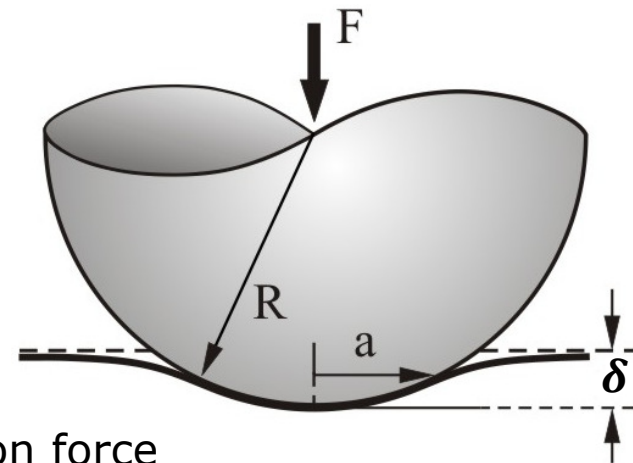
F is the force (from force curve)

E is Young's modulus (fit parameter)

ν is Poisson's ratio (typically 0.2-0.5)

R is the radius of the indenter (tip)

δ is indentation depth



- Model is valid for $a \ll R$
- DMT model: Hertzian model with adhesion force
- Linearized model is more robust, no need to find contact point
- **Question:** if we use a 5 μ m radius sphere probe to indent into sample surface by 50nm, which makes the contact area radius about 700nm, what is the tip radius **R** for above equation?

Sample Modulus: Sneddon Model



$$F = \frac{2}{\pi} \frac{E}{(1 - \nu^2)} \tan(\alpha) \delta^2 \quad \text{or} \quad (F)^{1/2} = \left(\frac{2}{\pi} \frac{E}{(1 - \nu^2)} \tan(\alpha) \right)^{1/2} \delta$$

Linearized Equation

Where

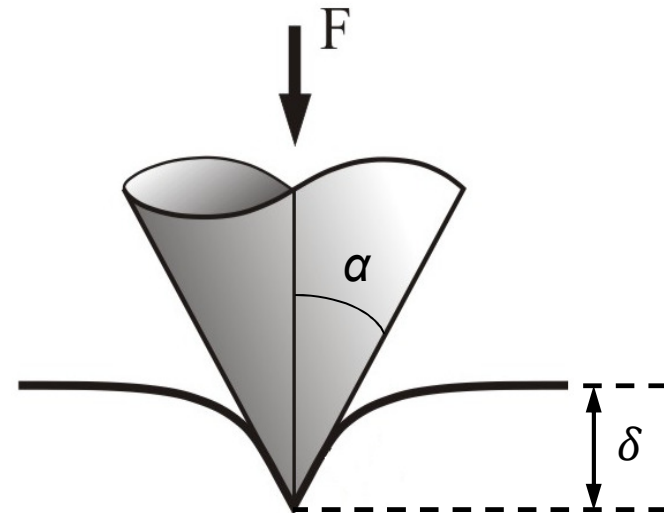
F is the force (from force curve)

E is Young's modulus (fit parameter)

ν is Poisson's ratio (typically 0.2-0.5)

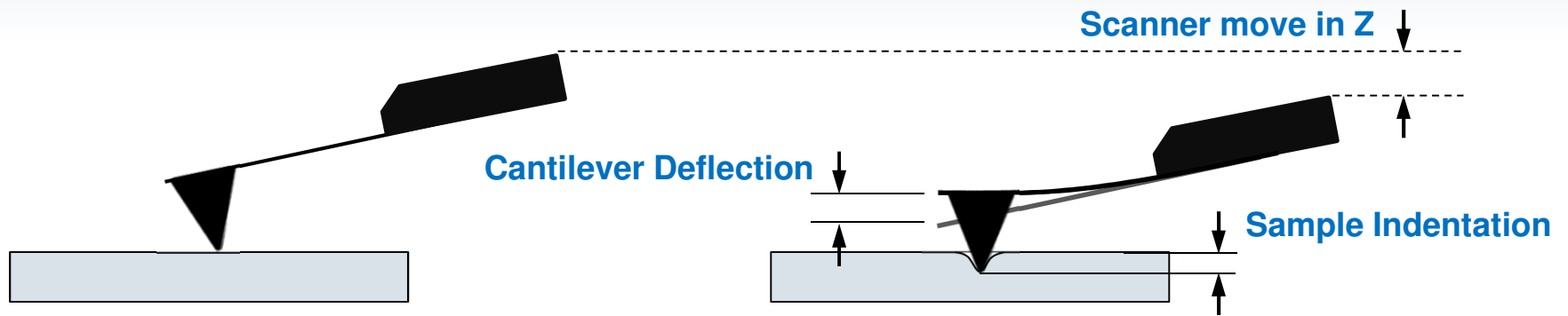
α is half angle of the indenter

δ is indentation depth



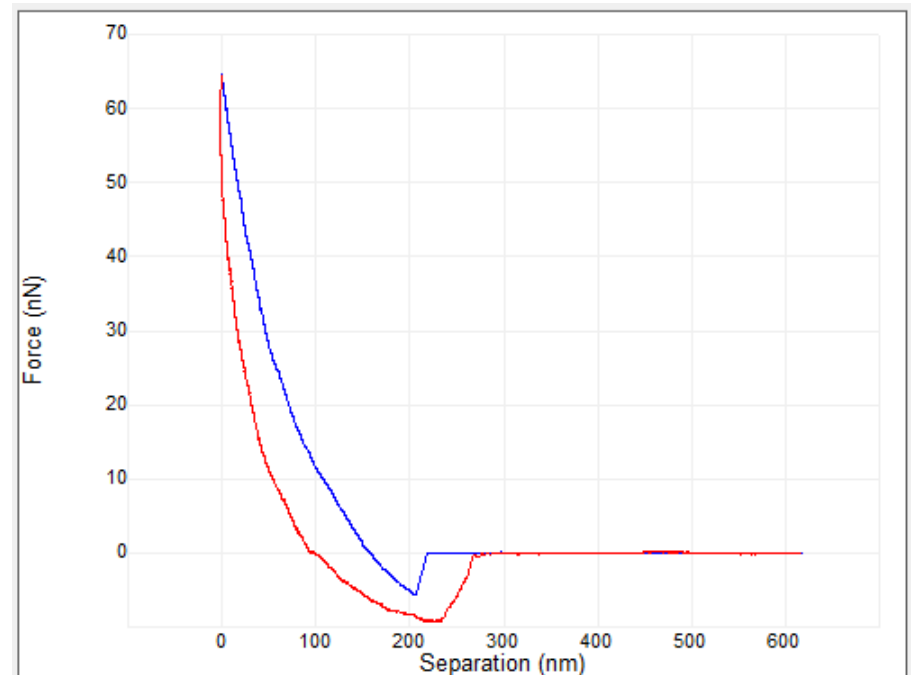
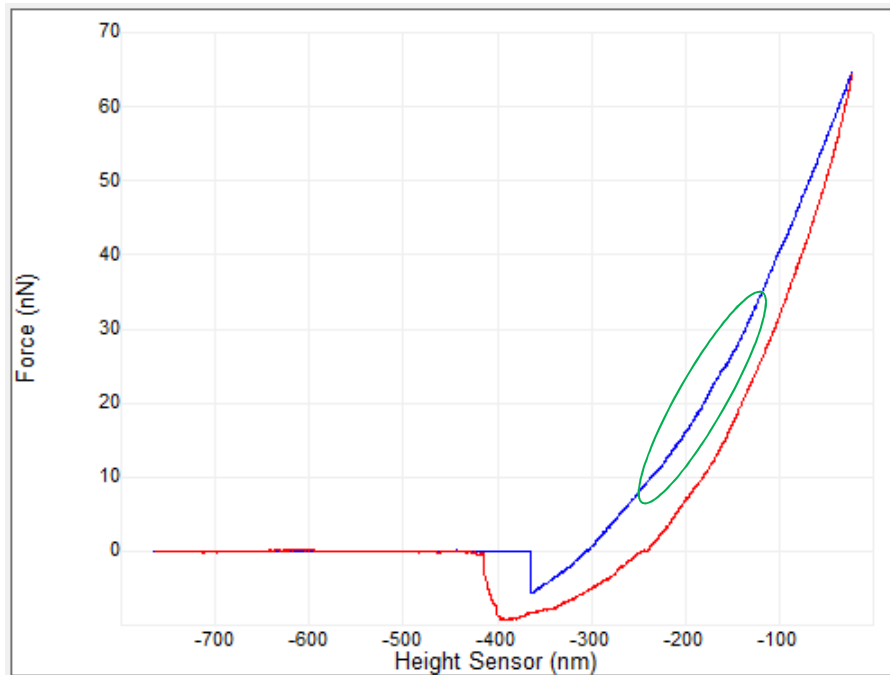
- Used for large sample indentation
- Uses approach curve, includes plastic deformation
- **Question:** is δ same as the Z scanner movement after tip contacts sample surface?

Indentation Depth



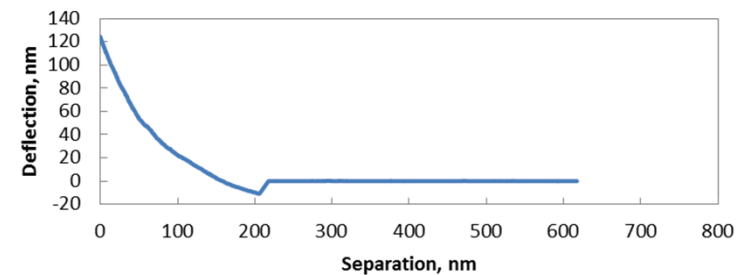
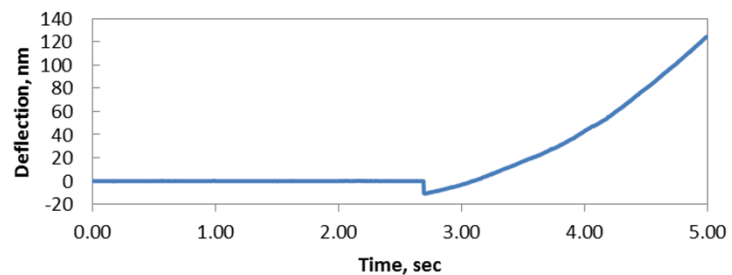
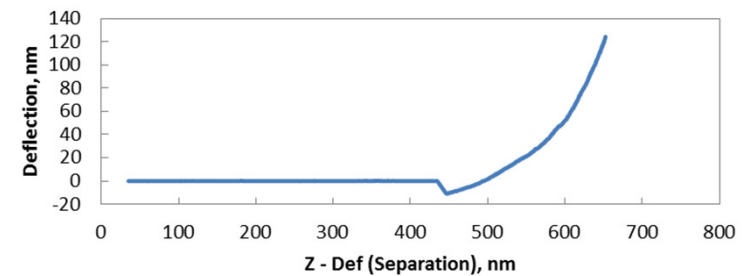
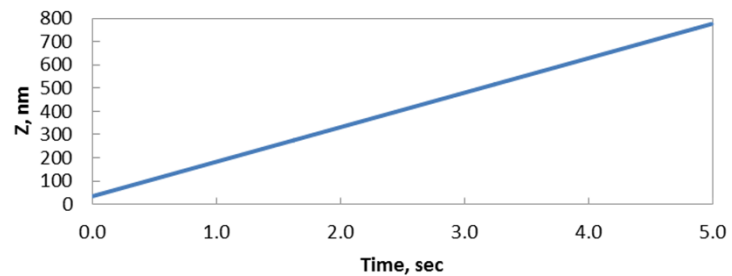
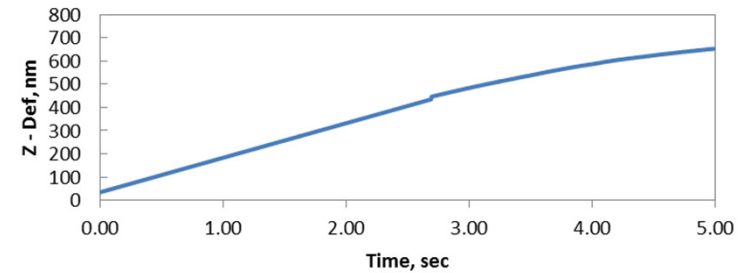
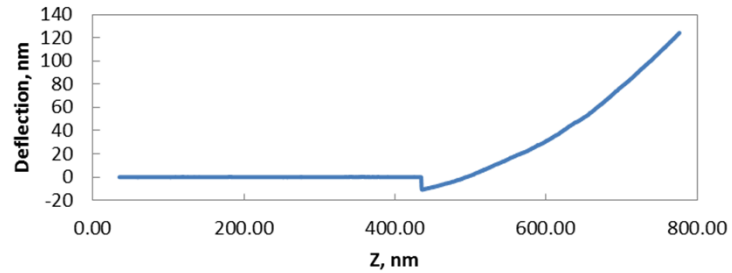
- During contact
 - Scanner movement in Z(↓) direction causes **cantilever deflection(↑)** and **sample indentation (↓)**
 - $Z = \text{Sample Indentation} + \text{Cantilever Deflection Change}$
- Separation = $Z - \text{Cantilever Deflection Change}$
 - Tip actual travel position with max indentation point as reference
- Indentation Depth: amount of Separation change after initial contact (**indentation depth should be always positive**)

Force-Distance vs. Force-Separation

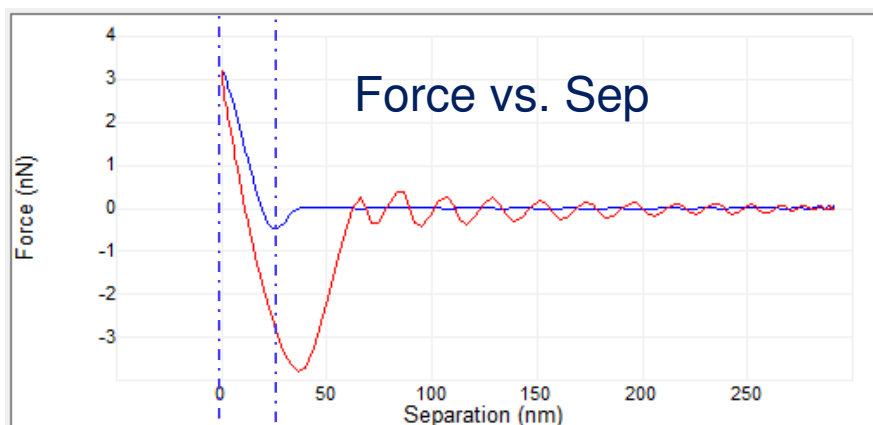
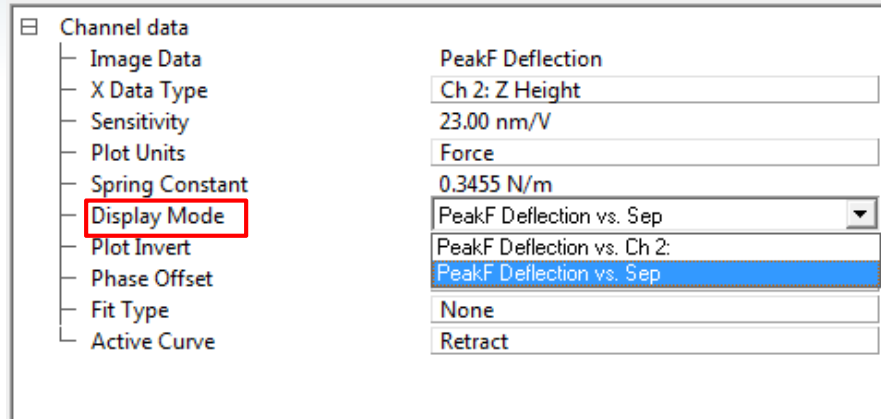
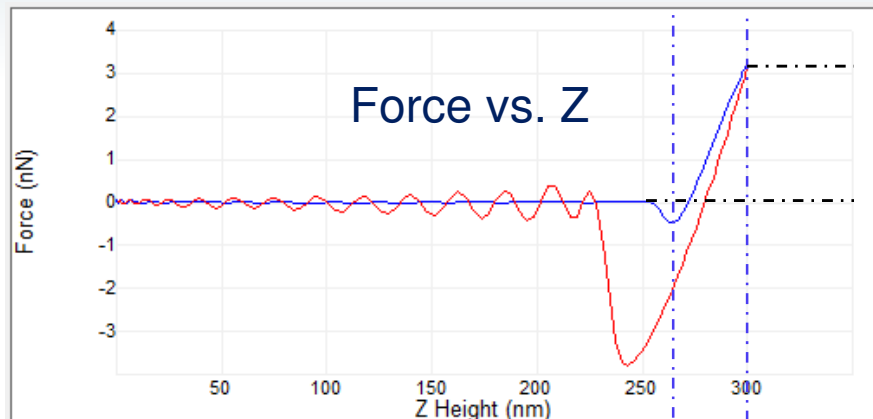


- Force-Distance curve needs to be converted to Force-Separation curve before fitting with the contact model equations

How to Convert Force Curves

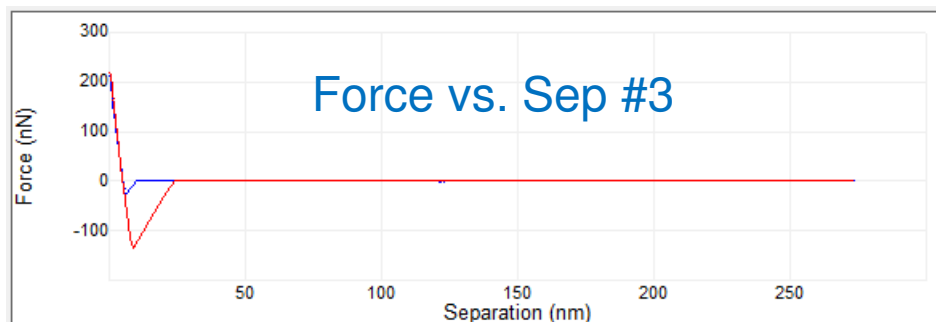
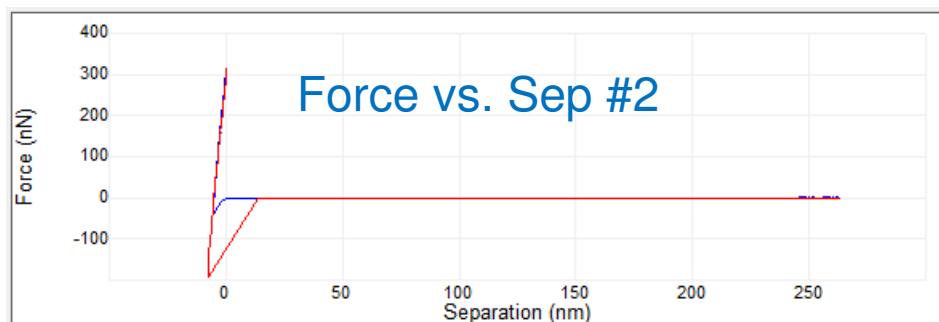
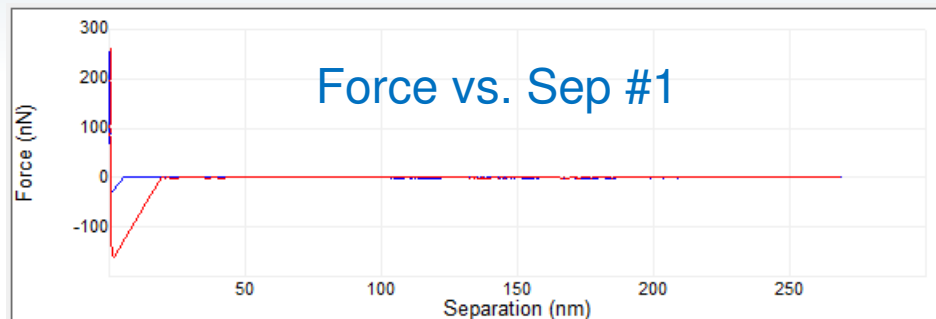
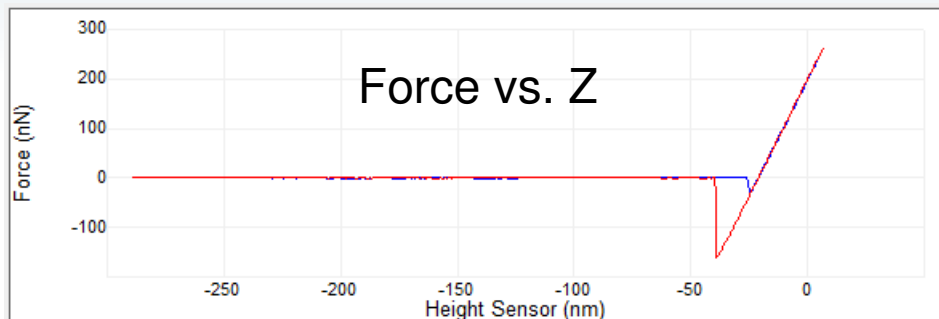


Force vs. Separation Curve



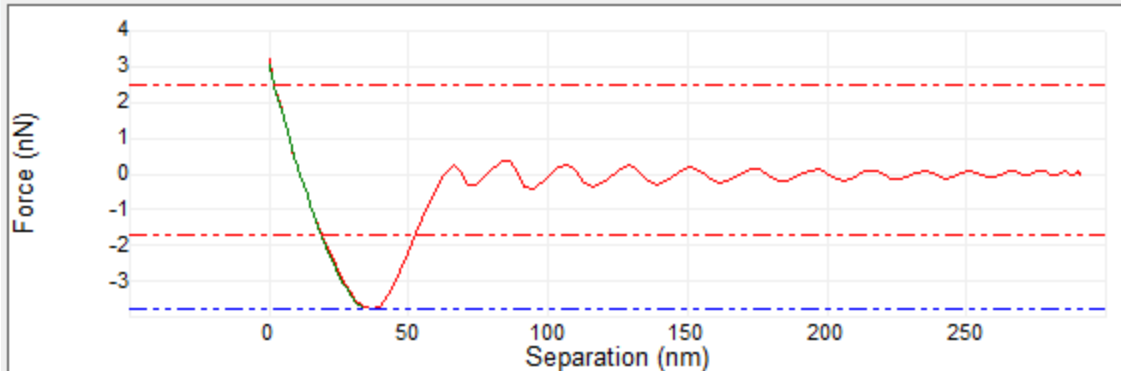
- Force vs. Z curve needs to be converted into Force vs. Sep curve in order to be fitted by the contact models
- This conversion can be easily done in Nanoscope Analysis software

Example separation curves



- The above **Force vs. Z** curve is obtained by a soft cantilever ramp on a very hard sample, e.g. a ScanAsyst-Air probe on Sapphire sample
- **Questions:** Which of the above **Force vs. Sep** curves is correct?

How to get modulus from force curve: DMT Model



Inputs	
Active Curve	Retract
Fit Method	Linearized Model
Include Adhesion Force	Yes
Max Force Fit Boundary	90 %
Min Force Fit Boundary	30 %
Fit Model	Hertzian (Spherical)

Results	
R ²	0.9998
Young's Modulus	3.64 MPa
Reduced Modulus	4.85 MPa

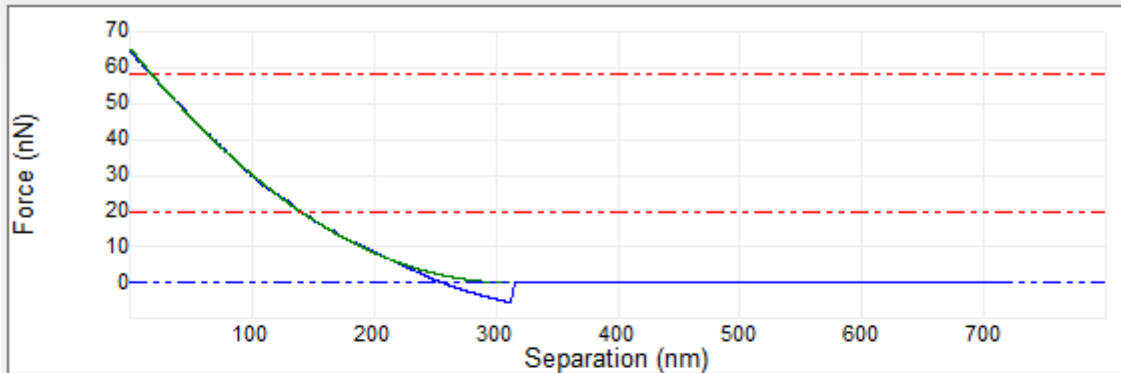
- Click  (Indentation) for force curve analysis
- DMT Model:
 - Use retract curve
 - Fit with linearized model
 - Include Adhesion Force:
 $F = F_{tip} - F_{adh}$
 - Use Hertzian (Spherical)
 - Set force fit region
- Young's Modulus E and Reduced Modulus E^* :

$$E^* = \left[\frac{1 - \nu_t^2}{E_t} + \frac{1 - \nu_s^2}{E_s} \right]^{-1}$$

$$\text{If } E_t \gg E_s$$

$$E^* = \frac{E_s}{1 - \nu_s^2}$$

How to get modulus from force curve: **Sneddon Model**



Inputs	
Active Curve	Extend
Fit Method	Linearized Model
Include Adhesion Force	No
Max Force Fit Boundary	90 %
Min Force Fit Boundary	30 %
Fit Model	Sneddon (Conical)

Results	
R ²	0.9996
Young's Modulus	2.98 MPa
Reduced Modulus	3.27 MPa

- Sneddon Model:
 - Indentation should $\gg$ tip radius
 - Use Extend curve
 - Fit with linearized model
 - Don't include Adhesion Force
 - Use Sneddon (Conical)
 - Set force fit region
 - Typically used for cell or gel in liquid

Cantilever Selection



Method_1

Sample Modulus (E)	Probe	Nominal Spring Constant (k)
1 MPa < E < 20 MPa	ScanAsyst-Air	0.5 N/m
5 MPa < E < 500 MPa	Tap150A, P/N MPP-12120-10	5 N/m
200 MPa < E < 2000 MPa	Tap300A (RTESPA), P/N MPP-11120-10	40 N/m
1 GPa < E < 20 GPa	Tap525A, P/N MPP-13120-10	200 N/m
10 GPa < E < 100 GPa	DNISP-HS	350 N/m

Method_2

$$F = \text{Deflection} * K = \frac{4}{3} \frac{E}{(1 - \nu^2)} \sqrt{R} \delta^{3/2}$$

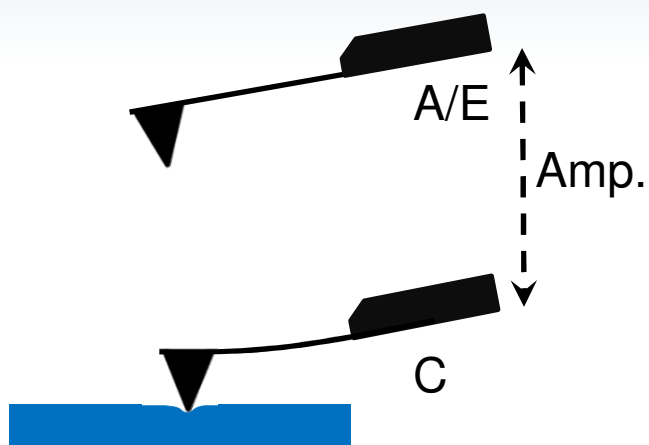
- Cantilever spring constant needs to match with the sample modulus
- For accurate modulus measurement, the amount of sample Indentation(at least 2-5nm) should be greater than cantilever deflection

Cantilever Deflection =< Sample Indentation

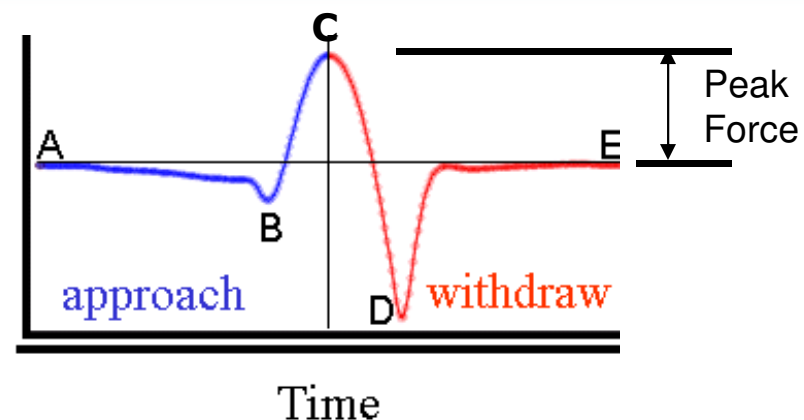
- **Question:** What spring constant to choose to measure 10MPa modulus with a 5um diameter sphere probe?

PF-QNM

Peak Force Tapping

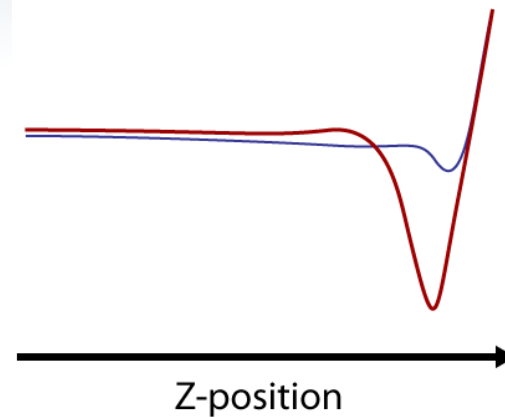
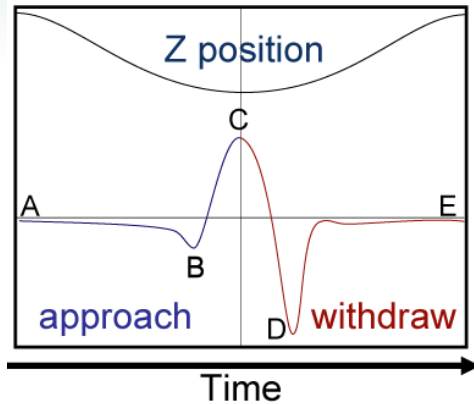


One cycle: photo detector vertical deflection (force) vs time.

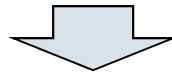


- A - Probe tip 300 nm above sample surface (150nm PFT amplitude). Z piezo pushes probe toward sample.
- B - Tip contacts sample ("snap-to-contact"). Z piezo pushes probe further until reach PeakForce.
- C - After PeakForce position, probe starts withdrawing from sample.
- D - Probe tip breaks free off sample at maximum adhesion point.
- E - Probe back to starting point 300 nm above sample surface
- **Probe repeat the cycle A->E cycle at fixed frequency, typically 2KHz**

Difference Between PF-Tapping and FD Curve

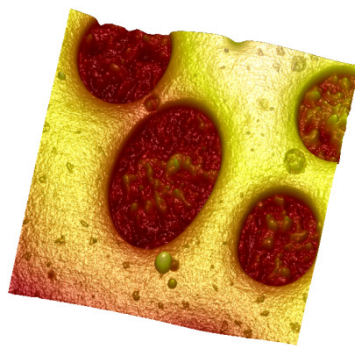
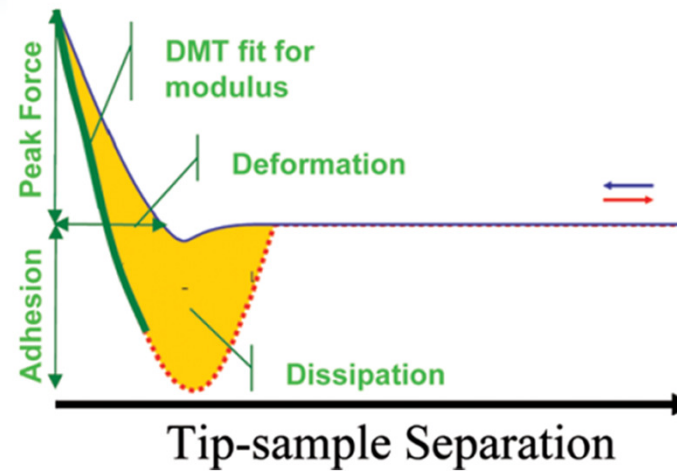


- Sine wave drive signal, 2KHz vs. triangular wave drive signal, ~2Hz
- Sync distance + feedback vs. trigger threshold
- Real-time background subtraction vs. offline baseline correction
- Advanced real-time data analysis vs. offline analysis

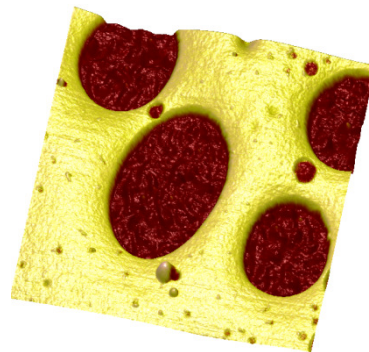


- Faster speed and higher resolution image
- Accurate force control and gentler tapping
- Real time high resolution mechanical properties mapping

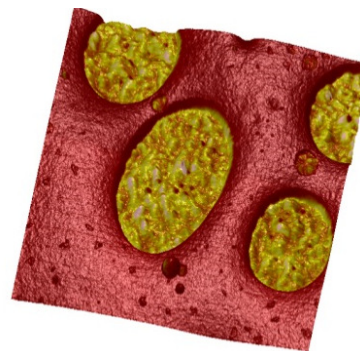
PeakForce QNM Data Channels



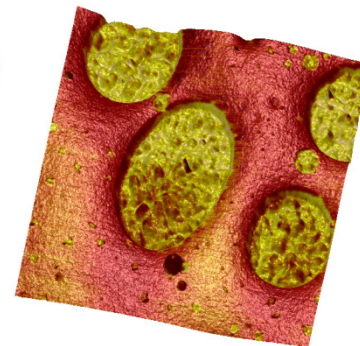
Height



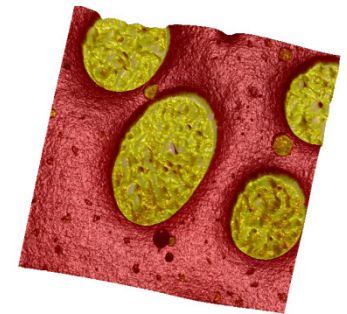
Modulus



Deformation

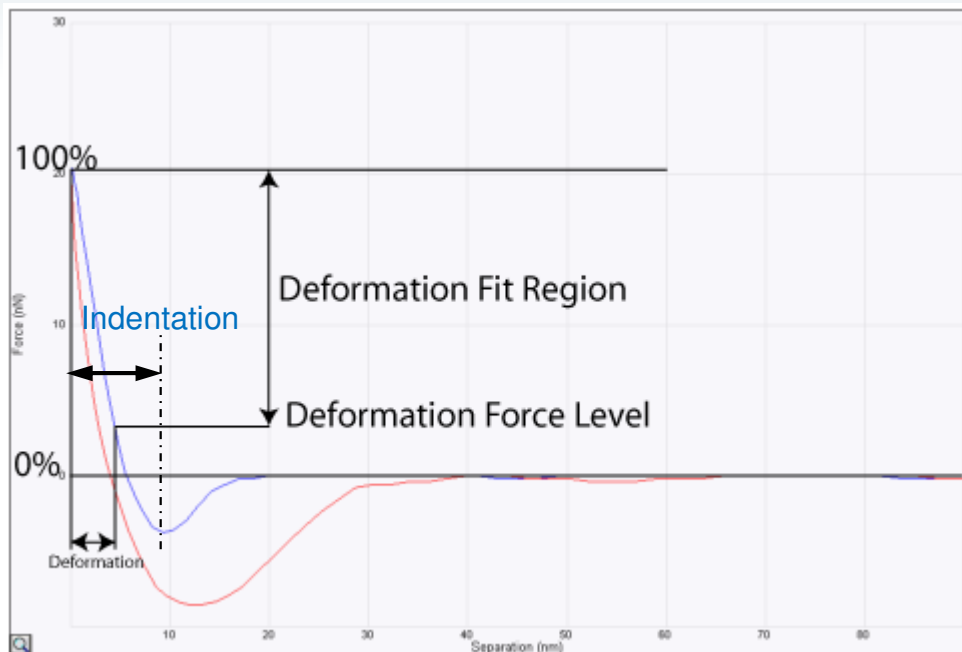


Adhesion



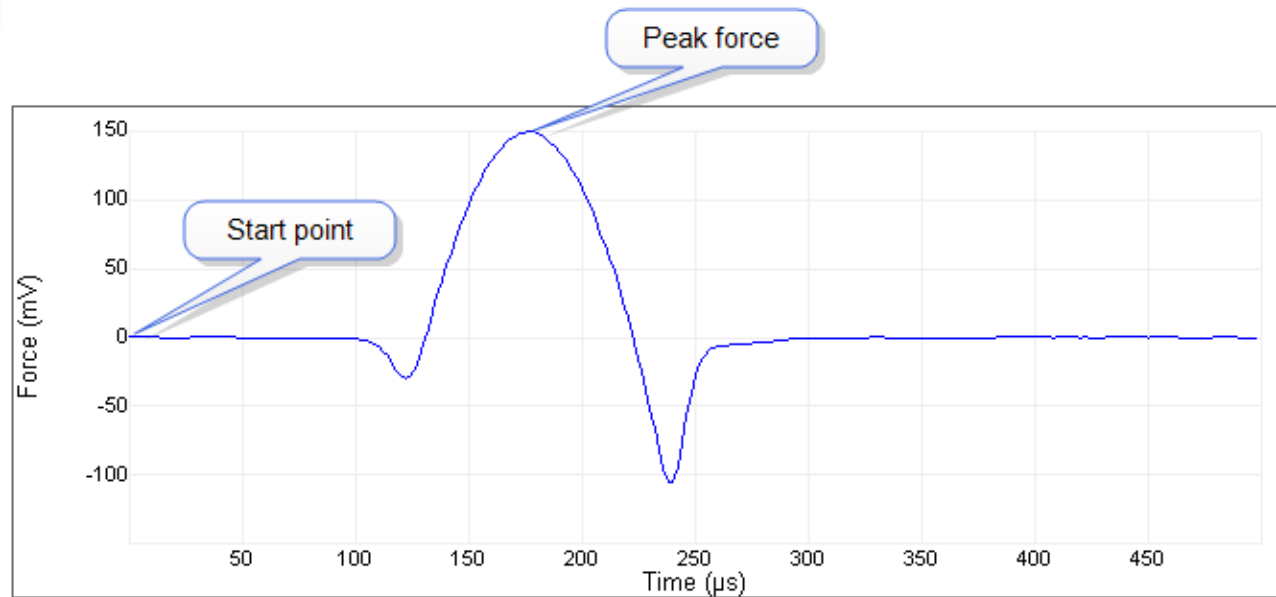
Dissipation

Deformation



- Softer sample has higher deformation
- Deformation calculation use approach force curve
- Deformation calculation generally underestimates sample indentation
- Since Deformation is based on Separation plot, it needs accurate Deflection Sensitivity

PFT Parameters: 1. Sync Distance

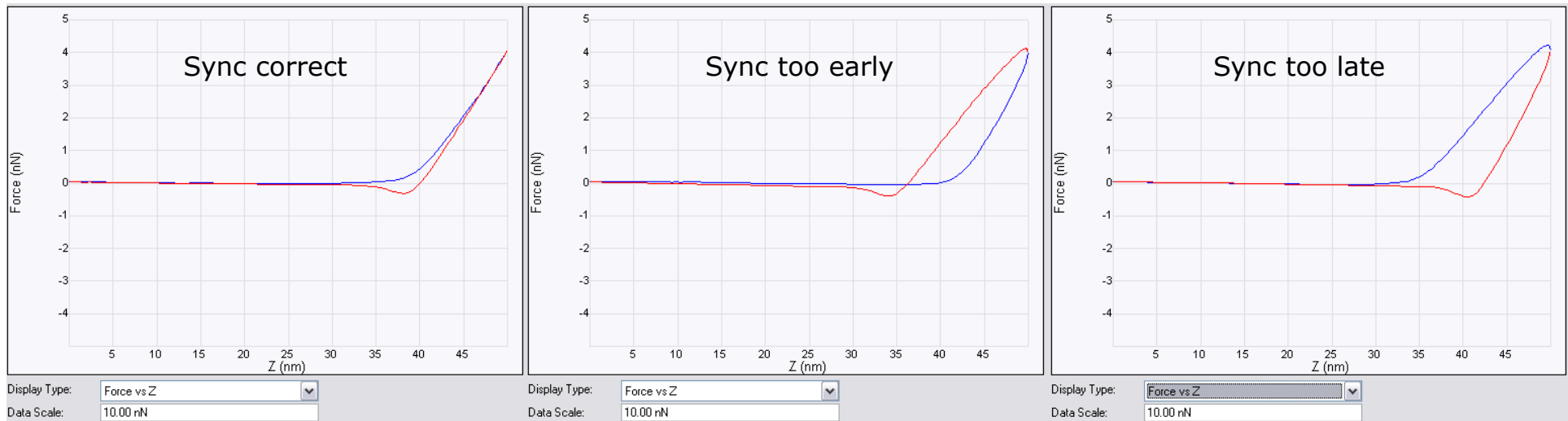


- Sync Distance is a time constant during each tapping cycle when cantilever reaches the lowest position.
- On hard sample deflection reaches max when piezo extends to lowest position.
- In QNM analysis, Sync Distance or PeakForce position is used as the turning point between approach force curve and retract force curve.

Typical Sync Distance Problems



- Two common “Sync” problems
 - Late sync- approach curve contains decreasing force before turnaround. Might not be a problem – could be due to sample viscoelasticity
 - Early sync- retract curve contains increasing force at beginning of turnaround.
- Use Auto Config button on hard sample to measure sync distance



PFT Parameters: 2. PeakForce Setpoint



- Set ScanAsyst control to Individual

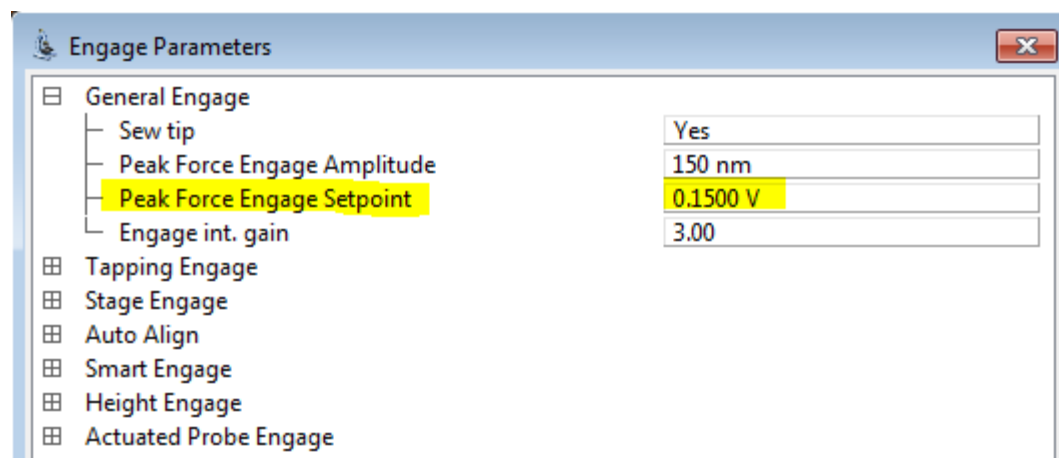
ScanAsyst Auto Control	Individual
ScanAsyst Auto Gain	On
ScanAsyst Auto Setpoint	Off
ScanAsyst Auto Scan Rate	On

- OK to engage with all (setpoint, gain, scan rate, z limit) *On*
- After engaged and stable turn auto setpoint *Off*
 - Use setpoint to control Indentation Depth
 - Effective tip radius depends on Indentation Depth
- Adjust the setpoint to have at least 2-5nm indentation for DMT model. For Sneddon model, Indentation should be larger, i.e. >30nm

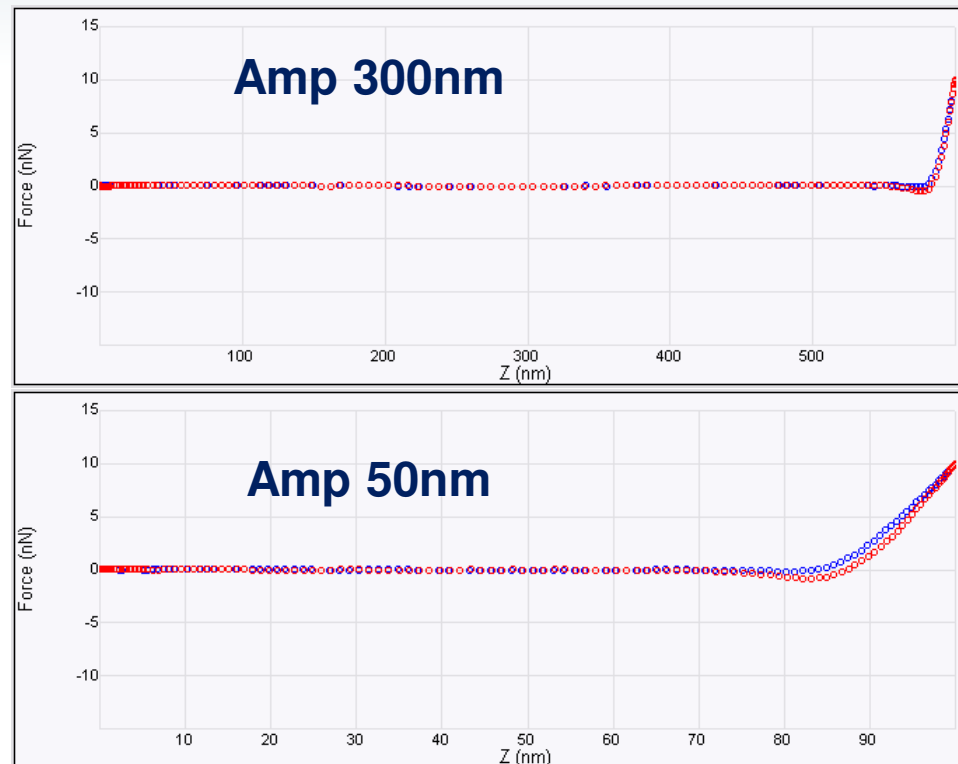
PFT Parameters: 3. Engage Force



- Engage ignores Peak Force Setpoint
 - Doesn't matter if ScanAsyst Auto Setpoint is On or Off.
- For engaging the Peak Force Engage Setpoint (in other controls) is used.
 - Default .15 V
 - OK for soft probe, e.g. ScanAsyst Air
 - Try .05 V for stiff probes, e.g. RTESPA

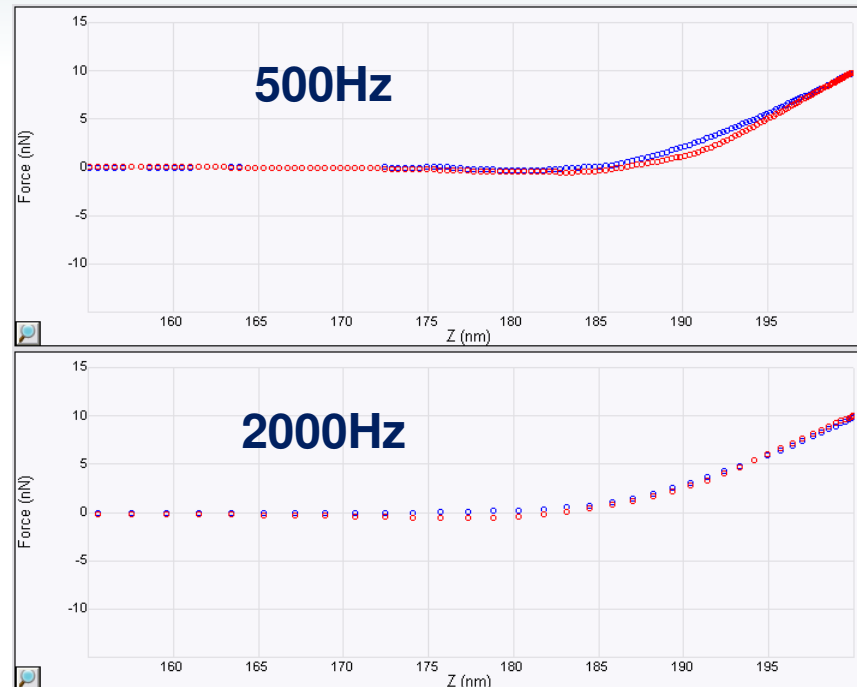


PFT Parameters: 4. Amplitude



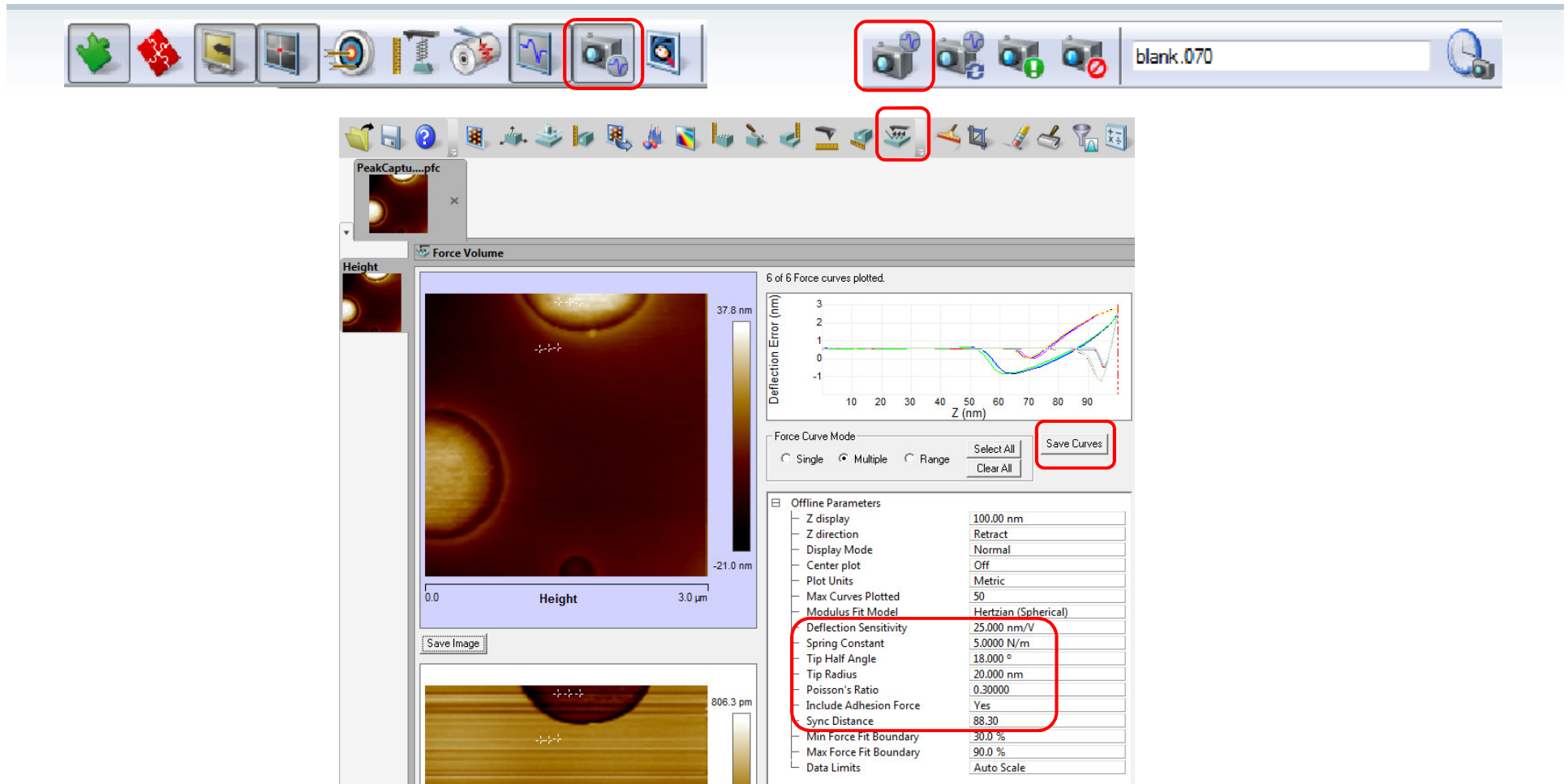
- Amplitude combined with setpoint determines the contact time ratio
- Recommended contact time 10-30%
- Higher contact time ratio means more data point in contact region
- Higher amplitude sometime is needed to overcome large adhesion force

PFT Parameters: 5. Frequency



- Default for Icon and MM is 2KHz, MM HR and Fast Scan is 8KHz
- Higher frequency can scan faster and/or higher resolution for topography
- Lower frequency means more data points in each force
- Lower frequency is preferred for fluid imaging, less hydrodynamic background
- DSP analyzes force curve at 1KHz with current software (V8.15, V9.0, and V9.1)

Peak Force Capture



- One force curve at each image pixel can be captured with PeakForce Capture
- Calibration parameters can be re-adjusted to data channels, e.g. modulus

Upcoming AFM Training Classes



AFM Courses in Santa Barbara, CA

Course	Date	Description
Fluid Imaging	March 10-12, 2015	3-Day course covering Image Quality, Fluid Imaging, and Image Processing
AFM Electrical Measurements	June 2-4, 2015	3-Day course covering KPFM, SCM & TUNA, and Piezo Force Microscopy
AFM Biological Applications	September 08-10, 2015	3-Day course covering DNA & Live Cell Imaging, PF-QNM on Bio Samples, and Force Spectroscopy
Mechanical Property Measurements	November 10-12, 2015	3-Day course covering Image Quality, Peak Force-QNM, and Heater Cooler

AFM Online Training Classes

Date	Time	Topic
Feb 3, 2015	11AM PST	PF-QNM I: Force Spectroscopy Register Here
Feb 17, 2015	11AM PST	PF-QNM II: Quantitative Nanomechanical Mapping Calibration Register Here
Mar 31, 2015	11AM PST	KPFM I: Amplitude Modulation
May 19, 2015	11AM PST	KPFM II: Frequency Modulation
Jul 7, 2015	11AM PST	Image Quality
Aug 11, 2015	11AM PST	Image Processing
Oct 6, 2015	11AM PST	TUNA/PF-TUNA
Nov 24, 2015	11AM PST	Scanning Capacitance Microscopy

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Your Cost Of Downtime



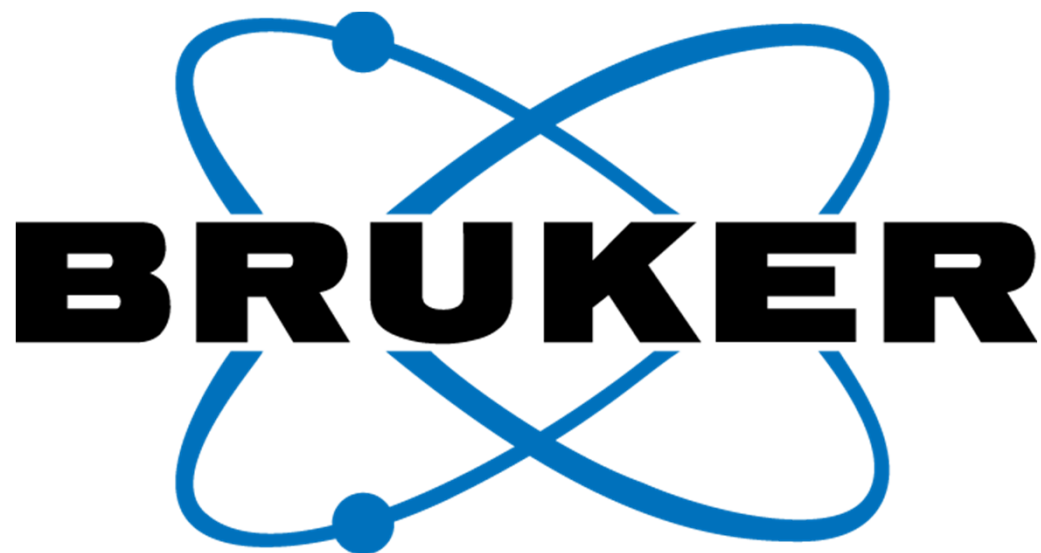
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